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Hardware

Portenta H7



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Portenta H7 as a Wi-Fi Access Point

In this tutorial you will configure the Portenta H7 as an access point and build a simple web server that will allow you to control the built-in RGB LEDs from your mobile device.

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Overview

Portenta H7 comes with an on-board Wi-Fi and a Bluetooth® Module that allows to develop IoT applications that require wireless connectivity and Internet access. Turning the board into an access point allows to create a Wi-Fi network on its own and allows other devices to connect to it. In this tutorial you will learn to set up your board as an access point web server and remotely control the red, green and blue LEDs on the built-in RGB LED by accessing an HTML page on your mobile device's browser.

Goals

About the built-in Wi-Fi + Bluetooth® module.

How a client-server model works

How to create an HTTP

[Help](#)

Required Hardware and Software

[Portenta H7 \(ABX00042\)](#) or
[Portenta H7 Lite
Connected \(ABX00046\)](#)

One USB-C® cable (either
USB-A to USB-C® or USB-
C® to USB-C®)

Arduino IDE 1.8.10+ or
Arduino Pro IDE 0.0.4 +

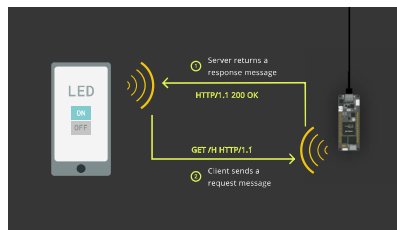
A smart phone

Access Point Configuration

The Portenta H7 features a [Murata 1DX](#), which is a high performance chipset which supports Wi-Fi 802.11b/g/n + Bluetooth® 5.1 BR/EDR/LE, up to 65 Mbps PHY data rate on Wi-Fi and 3 Mbps PHY data rate on Bluetooth®. This module helps to configure the Portenta into three different modes of operation - an Access Point, a Station, or both. In this tutorial we will only focus on the access point configuration.

When the board is configured to operate as an access point, it can create its own wireless LAN (WLAN) network. In this mode, the board transmits and receives signals at 2.4 GHz allowing other electronic devices with Wi-Fi capabilities using the same bandwidth to connect to the board

With the access point set up, you create a client server architecture where the board provides a web server communicating with the client devices over HTTP. The connected devices can then make HTTP GET requests to the server to retrieve web pages served by the web server on the board. This makes the Portenta H7 an ideal board for developing IoT solutions where external client devices can send and receive information while more complex processing tasks take place on the server.

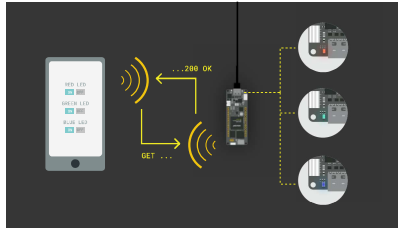


Instructions

Setting Up the Web Server

In this tutorial you are going to convert the board into an access point and use it to set up a web server which provides a HTML webpage. This page contains buttons to toggle the red, green and blue color of the built-in LED. You will then connect your mobile device to this access point and access this web page through the browser on your mobile phone. Once retrieved, you will be able to control the state of the red, green and blue

LED on the built-in RGB LED from your mobile device.



1. The Basic Setup

Begin by plugging in your Portenta board to your computer using a USB-C® cable and open the Arduino IDE. If this is your first time running Arduino sketch files on the board, we suggest you check out how to [set up the Portenta H7 for Arduino](#) before you proceed.



The Portenta H7 can be connected to the computer using an appropriate USB-C® cable

2. Create the Web Server Sketch

Next you need to create a web server sketch that will handle the HTTP GET requests and provide the client devices with the HTML web page. The [Wi-Fi](#) library provides all necessary methods that allows Arduino boards to use their Wi-Fi features provided by

following code, paste it into a new sketch file and name it **SimpleWebServer.ino**.

Note: You can access the final sketch inside the library:

Examples >

Arduino_Pro_Tutorials >

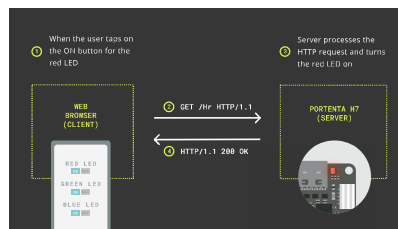
Portenta H7 as a Wi-Fi Access

Point > SimpleWebServer

```
1  #include <WiFi.h>
2  #include "arduino_s
3
4  //please enter
5  char ssid[] = SECRE
6  char pass[] = SECRE
7  int keyIndex = 0;
8
9  int status = WL_IDLI
10
11 WiFiServer server(80
12
13 void setup() {
14     // put your setup
15     Serial.begin(9600
16     while (!Serial) {
17         ; // wait for se
18     }
19
20     Serial.println("Ar
21
22     pinMode(LED_R, OUTP
23     pinMode(LED_G, OUTP
24     pinMode(LED_B, OUTP
25
26     // by default the
27     // you can overrie
28     // WiFi.config(IP,
```

This sketch describes how the server will handle an incoming HTTP GET request from a client, both to request the HTML web page from the server and the requests to change the LED states using dedicated URLs

Here the web page is just a simple HTML page with buttons to toggle the LED states. The way in which the web page works is: whenever a button on the web page is pressed, the client device (in this case your phone) sends a HTTP GET request to a URL denoted by a letter, in this case H or L (H stands for HIGH, L stands for LOW) followed by the LED color that should be turned on or off r, g or b. For example, to turn on the red LED the URL is /Hr . Once the server receives this request, it changes the corresponding LED state, closes the connection and continues to listen to next requests.

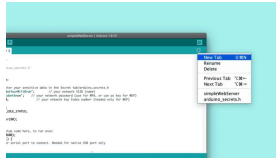


Remember that the built-in RGB LEDs on the Portenta H7 need to be pulled to ground to make them light up. This means that a voltage level of **LOW** on each of their pins will turn the specific color of the LED on, a voltage level of **HIGH** will turn them off.

3. Create the arduino_secrets.h Tab

A good practice is to have sensitive data like the SSID and the password required to identify and connect to a certain network.

arrow icon below the Serial Monitor button and open a new tab in the Arduino IDE. This will create a new file.



Open a new tab in the IDE

Name the file **arduino_secrets.h** and click OK.



Naming the new tab
arduino_secrets.h in
the IDE

Once you have created the new tab, you will see an empty page in the IDE. Define two constants `SECRET_SSID` and `SECRET_PASS` that will hold the name of the Wi-Fi network and the corresponding password. Add the following lines to your **arduino_secrets.h** file:

```
1 # define SECRET_SSID "  
2 # define SECRET_PASS "
```

The SSID (PortentaAccessPoint) and password (123Qwerty) are placeholder strings made for this tutorial. For

i security reasons you should rename them to something memorable but not easy to guess. Keep in mind that the password must be at least 8 characters long.

In order to access the

`SECRET_SSID` and `SECRET_PASS`

constants in the

simpleWebServer.ino sketch

file, the header file that you have

just created needs to be

included. In your sketch file this

has already been taken care of by

the following line at the

beginning of the sketch:

```
1 #include "arduino_secrets.h"
```

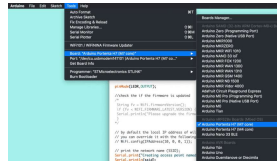


Including the header file `arduino_secrets.h` in the sketch file

4. Upload the Code

Select the **Arduino Portenta H7 (M7 core)** from the **Board** menu and the port the Portenta is connected to. Upload the

Doing so will automatically compile the sketch beforehand.

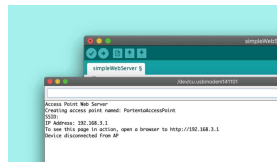


Uploading the
SimpleWebServer.ino
to the Portenta

Once you have uploaded the code, open the Serial Monitor. You will be able to see the IP address of the access point. You will also see the message,

Device disconnected from AP

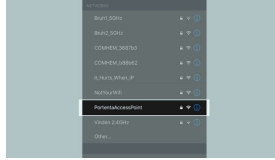
which means there are no devices connected to the Access point yet.



Serial monitor
displaying the details
of the Access point

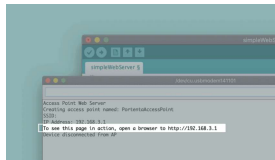
5. Connecting to the Portenta Access Point

Once the access point is active and ready to be connected with external devices, you will be able to find the **PortentaAccessPoint** on the list of networks on your mobile device. Once you have entered the password you have defined earlier, your smart phone will connect to access point.



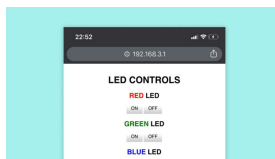
PortentaAccessPoint
shown on the list of
available network
devices

Now open a browser window on
your mobile device and copy &
paste the URL containing
Portenta's IP address that is
displayed on the serial monitor.



The URL containing the
IP address of the
access point displayed
in the serial monitor

Once you have entered the URL,
the client sends a GET request to
the web server to fetch the HTML
web page specified in the code.
Once loaded, you will see the
web page in your mobile
browser.



The HTML web page
accessed on your
mobile browser

6. Access the Board From

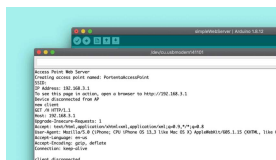
If you take a look at the Serial Monitor, you can see the details of the HTTP GET request and other details of the device connected to the access point. The GET request is always in the following format:

```
1 GET URL HTTP/1.1
```

The URL is a string of characters sent to the server, in this case /Hx (where x stands for the color of the LED). This request containing the URL is received on the server and it replies with the following HTTP/1.1 response indicating that the connection was successful:

```
1 HTTP/1.1 200 OK
```

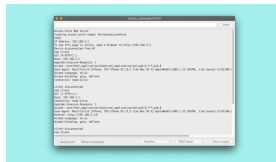
Once the server has responded to this request, it closes the connection and continues listening to next GET requests.



The client details displayed on the serial monitor

You are now be able to toggle the states of the red, green and blue LED through the buttons

button, the client sends a GET request to a URL in the format /Hx or /Lx ,where x can be 'r', 'g' or 'b', depending on the button pressed on the HTML page. The web server then reads the URL requested by the client, changes the state of the LED corresponding to the URL and closes the connection.



The GET request details displayed in the serial monitor

Conclusion

This tutorial shows one of the several capabilities of the on-board WiFi+Bluetooth® module by configuring the board as an access point and setting up a web server. You have also learnt how a simple client-server model and the underlying HTTP requests and responses work.

Next Steps

Now that you have learnt how to set up a board as an access point and understand the client-server model, start experimenting with the **simpleWebServer.ino** sketch. This sketch can be tweaked in a variety of ways based on your needs. For example, you can add a slider to

blink rate of the built-in RGB LED from your mobile device.

Suggest changes

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